

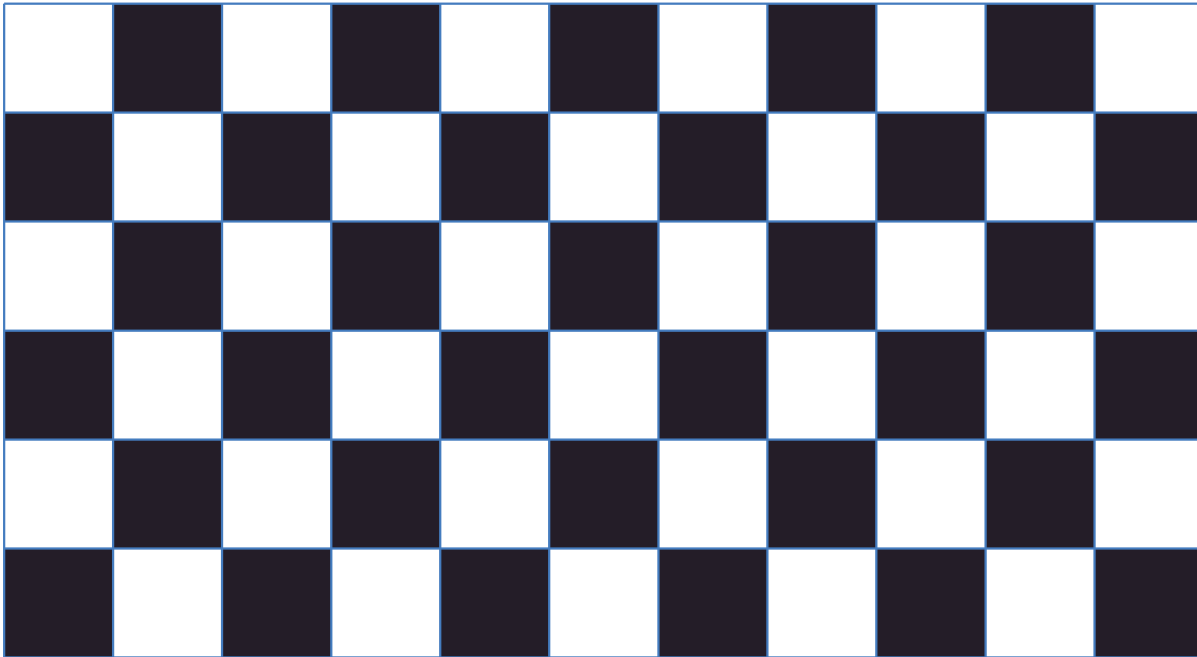
Description:

Checkerboard Resolution

In this kata we will be counting the the black squares on a special checkerboard. It is special because it has a `resolution` which determines how the black and white squares are laid out.

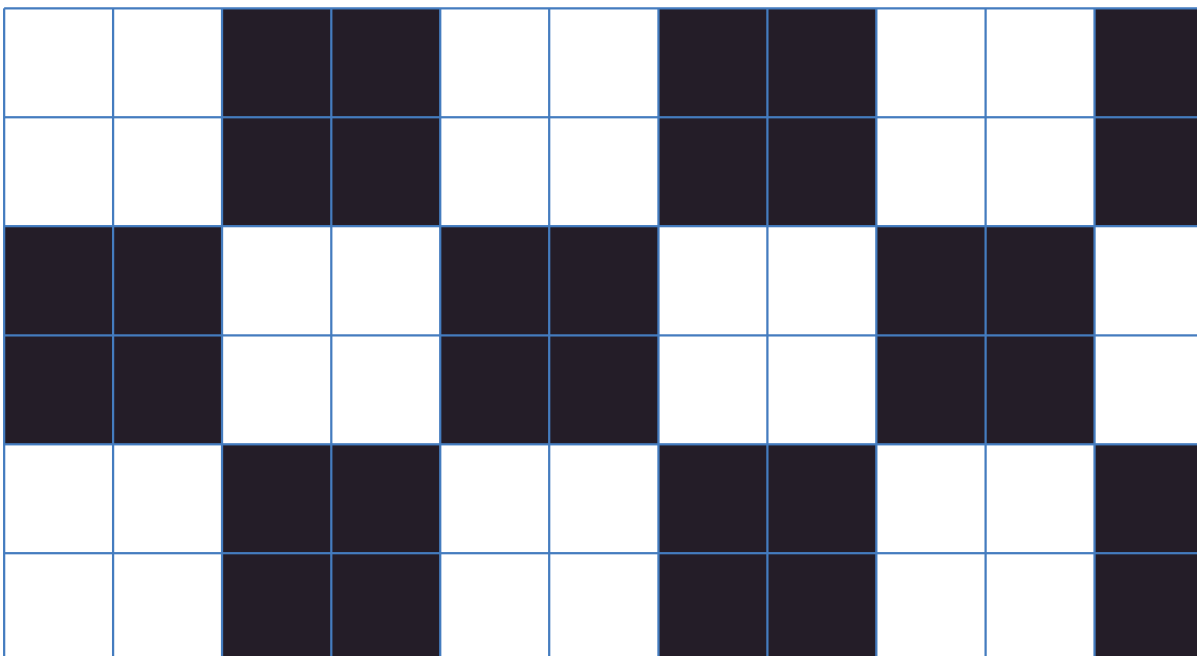
The `resolution` refers to the dimensions of squares of a single colour. See below for an example with dimensions 11x6:

With `resolution = 1`:



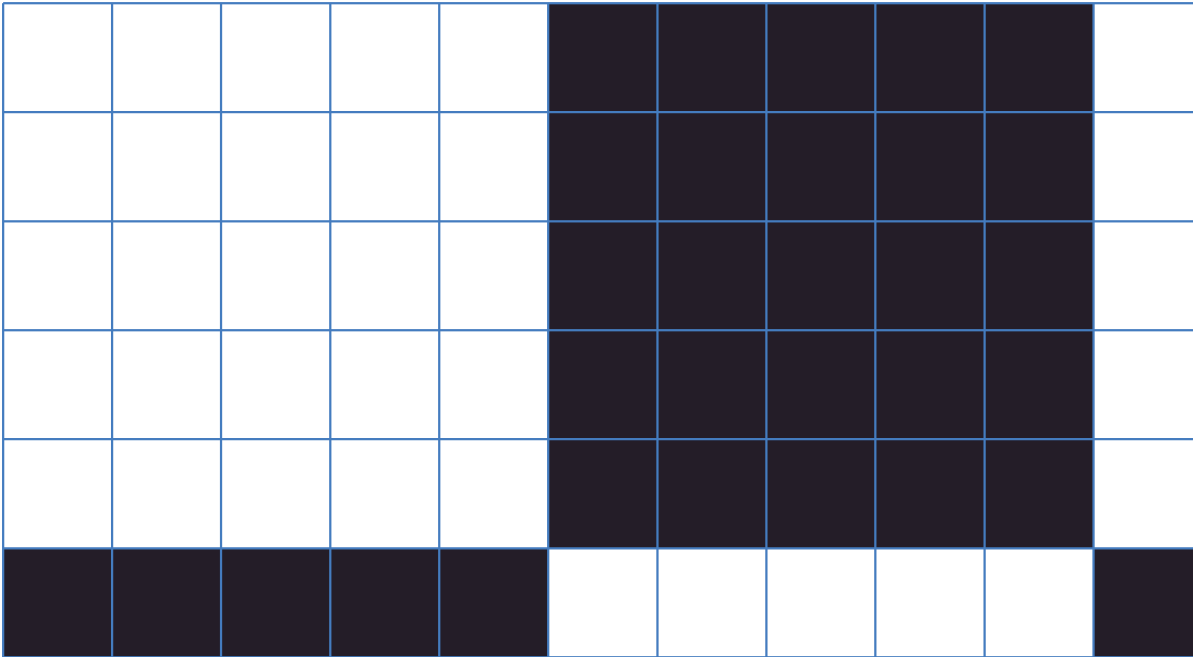
Number of black squares = 33

And now with `resolution = 2`:



Number of black squares = 32

And one more example, resolution = 5:



Number of black squares = 31

Credit to awesomead for the pretty images!

As you may have noticed the top left square is **always white**, and we are counting the **individual** black squares on the board.

Task

You are required to write a function that will take in three parameters:

- width -> The width of the board
- height -> The height of the board
- resolution -> The size of the coloured squares on the checkerboard (as shown above)

And returns the total count of all individual black squares.

Additional Info

- All inputs will be valid
- $0 < \text{width} \leq 10^{32}$
- $0 < \text{height} \leq 10^{32}$
- $1 \leq \text{resolution} \leq 10^{32}$

Good luck!

Performance

Algorithms

Matrix